

Computer Vision Notes

1. Object Detection

Object detection is a computer vision technique used to identify and locate objects in an image or video. It performs both classification and localization by recognizing the object type and drawing bounding boxes around detected objects.

- Detects multiple objects in an image
- Provides object label and location
- Methods: Haar Cascade, HOG + SVM, YOLO, SSD, Faster R-CNN
- Applications: Autonomous vehicles, surveillance systems, robotics, medical imaging

2. Face Detection

Face detection is the process of detecting the presence and position of human faces in images or videos. The system searches for facial features such as eyes, nose, mouth, and facial symmetry.

- Detects human faces only
- Viola-Jones algorithm uses Haar features, AdaBoost, and cascade classifiers
- Modern methods include CNN-based detectors and MTCNN
- Applications: Face unlock, attendance systems, CCTV surveillance

3. Pedestrian Detection

Pedestrian detection identifies humans in images or video streams and is mainly used in transportation and safety systems.

- Uses body shape, contours, and motion information
- Methods: HOG + SVM, YOLO, Faster R-CNN
- Challenges: Occlusion, poor lighting, crowded scenes
- Applications: Self-driving cars, crowd monitoring, accident prevention

4. Face Recognition

Face recognition is a biometric technique used to identify or verify a person using unique facial features.

- Stages: Face detection, feature extraction, feature matching, identification
- Types: Verification and Identification
- Techniques: Eigenfaces, Fisherfaces, LBPH, CNN, FaceNet
- Applications: Banking security, smartphone authentication, airport security

5. 3D Shape Models

3D shape models represent objects in three-dimensional space using depth information along with height and width.

- Components: Vertices, edges, surfaces, depth coordinates
- Types: Wireframe, surface, and solid models
- Reconstruction methods: Stereo vision, SfM, depth sensing
- Applications: Robotics, VR, gaming, medical imaging

6. Personal Photo Collections

Computer vision helps organize and manage large collections of personal photographs automatically.

- Functions: Face tagging, event grouping, duplicate detection
- Advantages: Easy searching and automatic album creation
- Applications: Google Photos, Apple Photos, Facebook tagging

7. Instance Recognition

Instance recognition identifies a specific object instance rather than just its category.

- Detects key points and extracts local features
- Methods: SIFT, SURF, ORB
- Applications: Product identification, landmark recognition, image retrieval

8. Category Recognition

Category recognition classifies objects into predefined categories such as car, dog, chair, or tree.

- Recognizes object category but not exact instance
- Methods: CNN classifiers, SVM classifiers, transfer learning
- Applications: Smart cameras, image search engines, autonomous systems

9. Context and Scene Understanding

Context and scene understanding analyzes the complete environment in an image and understands relationships between objects.

- Identifies scene type and object relationships
- Uses contextual information for improved recognition
- Techniques: Semantic segmentation, deep neural networks
- Applications: Autonomous driving, robotics, smart surveillance